Kathryn A. Mummah, Ph.D.

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Education

University of Wisconsin–Madison *Madison, WI*
PhD in Nuclear Engineering and Engineering Physics *May 2025*
- Dissertation: State-Level Nuclear Fuel Cycle Simulations for International Safeguards Applications
- Advisor: Professor Paul P.H. Wilson

University of Wisconsin–Madison *Madison, WI*
MS in Nuclear Engineering and Engineering Physics *May 2020*
- Graduate Certificates in Science Communication (LSC) and Energy Analysis & Policy (EAP)

University of Illinois at Urbana-Champaign *Urbana, IL*
BS in Nuclear, Plasma, and Radiological Engineering *May 2017*
- Concentration in Power, Safety, and the Environment
- Minor in Atmospheric Sciences

Experience

R&D Engineer, NEN-5 Nuclear Systems *Los Alamos, NM*
Los Alamos National Laboratory *Mar 2025 – present*
- Nuclear fuel cycle systems modeling for international nuclear safeguards, proliferation resistance, and nuclear material control & accounting,
- Special purpose reactor design and critical experiment modeling with MCNP
- Sponsors include: NNSA NA-20, NA-80, NRC, DOD/DOW, LDRD, industry partners

PhD student, Computational Nuclear Engineering Research Group *Madison, WI*
University of Wisconsin–Madison Dept. of Nuclear Engineering & Eng. Physics *Sept 2017 – Feb 2025*
- Advanced materials movement modeling fidelity within Cyclus to enable safeguards and MC&A analyses
- Reviewed communication efforts in consent-based siting of radioactive waste disposal facilities
- Designed uranium enrichment cascades in Cyclus to model post-JCPOA cascade performance scenarios
- Grader, NE405: Nuclear Reactor Theory, Spring 2018
- Sponsors: NNSA NA-22 (university consortia), NA-241

Graduate Research Assistant, NEN-1 Safeguards Science & Technology *Los Alamos, NM*
Los Alamos National Laboratory *Jan 2021 – May 2024*
- Developed tool to convert any Cyclus simulation into Code 10-style nuclear material accounting reports
- Generated synthetic advanced nuclear fuel cycles for use with nuclear fuel cycle simulators
- Sponsors: NNSA NA-241

Graduate Research Assistant, NEN-2 Advanced Nuclear Technology *Los Alamos, NM*
Los Alamos National Laboratory *May 2018 – Aug 2018*
- Conducted acquisition pathway analysis with APAT
- Taught nuclear engineering primer series including detailed background on nuclear energy industry
- Sponsor: Glenn T. Seaborg Institute

Visiting Student Guest, NEN-5 Systems Design & Analysis *Los Alamos, NM*
Los Alamos National Laboratory *May 2017 – Aug 2017*
- Assisted in MCNP deck development for nuclear thermal rocket concepts
- Created a reactor lab module to incorporate nonproliferation concepts into undergraduate curricula

Intern, Fuels Modeling & Simulations *Idaho Falls, ID*
Idaho National Laboratory *May 2016 – Aug 2016*
- Benchmarked pellet-cladding mechanical interactions in BISON

Undergraduate Research Assistant, Department of Nuclear, Plasma, and Radiological Engineering *Urbana, IL*
Aug 2015 – May 2017
University of Illinois at Urbana-Champaign

- Undergrad Research Assistant, Advanced Reactors and Fuel Cycle Research Group, 2016 – 2017
- Undergrad Research Assistant, Analysis of Reactor Transients & Stability Research Group, 2016
- Undergrad Research Assistant, Experimental Thermal Hydraulics Research Group, 2015

Intern, Reactor Engineering *Forked River, NJ*
Oyster Creek Generating Station, Exelon Generation *May 2015 – Aug 2015*

Intern, Spent Fuel & Decommissioning *Warrenville, IL*
Exelon Generation *May 2014 – Aug 2014*

Teaching Assistant, Department of Nuclear, Plasma, and Radiological Engineering *Urbana, IL*
Jan 2014 – Dec 2016
University of Illinois at Urbana-Champaign

- Lead Engineering Learning Assistant, ENG 100 Engineering Orientation Seminar for Dept. of Nuclear, Plasma, & Radiological Engineering. Fall 2015, Fall 2016
- NPRE 100: Introduction to Nuclear, Plasma, and Radiological Engineering. Fall 2014, Fall 2015, Fall 2016
- NPRE 101: Introduction to Energy Sources. Spring 2014, Fall 2015

Grader, Department of Astronomy *Urbana, IL*
University of Illinois at Urbana-Champaign *Aug 2014 – Dec 2014*

- ASTR 100: Introduction to Astronomy, Fall 2014

QuarkNet Summer Intern, Optical Astronomy *Batavia, IL*
Fermi National Accelerator Laboratory *June 2012 – Aug 2012*

Journals and Peer-reviewed Conferences

21. H. A. SMARTT, N. PAWLEY, L. PARKER, S. DORAWA, **K. MUMMAH**, M. RUCH, and Z. GASTELUM, "From Data to Insight: Multi-Modal Sensor Fusion for Efficient International Safeguards (DIMES)," in Proceedings of the IAEA Symposium on International Safeguards: Innovation and Resilience in a Changing World - Safeguards as a Shared Responsibility, International Atomic Energy Agency, Vienna, Austria (2026).
20. D. CHANDLER, B. PHATHANAPIROM, K. BRUNETT, R. CHAMBERLIN, S. CHIRYATH, L. CRABTREE, R. GUHA, D. HARTANTO, K. HOUGE, T. HANLON, N. HUBLEY, J. JOSHI, **K. MUMMAH**, M. PAUL, A. PRINCE, S. SKUTNIK, and L. WEEKS, "Conceptual Design of an Intrinsic Proliferation Resistance Optimization Methodology for Advanced Reactors and Associated Fuel Cycles," in Transactions of the American Nuclear Society Volume 135, 2026 ANS Winter Conference & Expo, Phoenix, AZ, USA (2026).
19. L. M. CRABTREE, M. PAUL, S. S. CHIRYATH, R. CHAMBERLIN, A. K. PRINCE, **K. A. MUMMAH**, K. K. HOGUE, S. E. SKUTNIK, N. HUBLEY, L. G. EVANS, J. JOSHI, T. E. HANLON, and D. CHANDLER, "Procedure for Using Intrinsic Proliferation Resistance Assessment Methodology to Evaluate Nuclear Reactor Feature Attributes of Advanced Reactors and Associated Fuel Cycles," in Transactions of the American Nuclear Society, 2026 ANS Winter Conference & Expo Volume 135, American Nuclear Society, Phoenix, AZ, USA (2026).
18. **K. A. MUMMAH** and P. P.H. WILSON, "A set of fictitious fuel cycle test cases for simulator demonstration and benchmarking," *Annals of Nuclear Energy* 238, (2026); https://doi.org/10.1016/j.anucene.2026.112503.
17. H. A. SMARTT, N. PAWLEY, L. PARKER, **K. MUMMAH**, M. RUCH, S. DORAWA, and Z. GASTELUM, "From Data to Insight: Multi-Modal Sensor Fusion for Efficient International Safeguards (DIMES)," presented at Institute for Nuclear Materials Management Annual Meeting, August 2026.
16. R. M. CHAMBERLIN, A. K. PRINCE, **K. A. MUMMAH**, N. H. HUBLEY, S. S. CHIRAYATH, K. K. HOGUE, *et al.*, "Methodology Development for Proliferation Resistance Optimization of Advanced Reactors and Fuel Cycles", presented at Pacific Basin Nuclear Conference 2026, Busan, South Korea (2026)

15. **K. A. MUMMAH** and P. P.H. WILSON, "A set of fictitious fuel cycle test cases for simulator demonstration and benchmarking," presented at Technical Workshop on Fuel Cycle Simulation (TWoFCS 2025), June 2025, Chicago, IL, USA, Argonne National Laboratory.
14. **K. A. MUMMAH** and P. P.H. WILSON, "Nuclear material packaging and flexible agent buying patterns in Cyclus," presented at Technical Workshop on Fuel Cycle Simulation (TWoFCS 2025), June 2025, Chicago, IL, USA, Argonne National Laboratory.
13. **K. A. MUMMAH** and P. P.H. Wilson, "Cyclus Toolkit Enhancements to Simulate Nuclear Material Buying Patterns," in Transactions of the American Nuclear Society, Annual Meeting 2024 ***130***, Las Vegas, NV (2024).
12. **K. A. MUMMAH**, "Bridging the Fidelity Gap in System-Scale Nuclear Fuel Cycle Simulations for Realistic State-Level Nuclear Material Accounting," presented at Advances in Nonproliferation Technology and Policy Conference 2023, November 2023, Washington, D.C. USA
11. **K. A. MUMMAH**, D. E. JACKSON, K. E. APT, J. OAKBERG, V. HENZL, and P. P.H. WILSON, "Simulated State accounting data for development of advanced and automated State-level safeguards analysis," Vienna, Austria (2023).
10. M. CALICE, D. BROSSARD, M. EDWARDS, S. GRUDER, **K. A. MUMMAH**, G. NEMET, S. WILLIAMS, and P. P.H. WILSON, "The Local Energy Transition: Perspectives on Community Energy Management in Wisconsin," in Proceedings of the Behavior, Energy and Climate Change (BECC) Conference, Washington D.C. (2022).
9. **K. A. MUMMAH**, "Learning from the 'Successful Failure' of a Low- and Intermediate-Level Radioactive Waste Project as a Case Study in Consent-Based Siting," presented at 2022 American Nuclear Society Student Conference, April 2022, Champaign, IL.
8. **K. A. MUMMAH** and P. P.H. WILSON, "Integrating Acquisition Pathway Analysis Into The Cyclus Fuel Cycle Simulator," in Proceedings of the 61st Annual Meeting of the Institute of Nuclear Materials Management, Jul. 2020, Virtual.
7. **K. A. MUMMAH** and P. P. H. WILSON, "Material Diversion Analysis Within a Fuel Cycle Simulator," presented at 2019 Technical Workshop of Fuel Cycle Simulation, June 2019, Champaign, IL.
6. **K. A. MUMMAH**, R. BAHRAN, K. MILLER, and P. P.H. WILSON, "Acquisition Pathway Analysis using Fuel Cycle Simulators," in Proceedings of the 60th Annual Meeting of the Institute of Nuclear Materials Management, Jul. 2019, Palm Desert, CA, USA.
5. B. MOUGINOT, **K. A. MUMMAH**, and P. P. H. WILSON, "Assessing Material Inventory Uncertainties in Integrated Fuel Cycle Simulations," presented at Advances in Nuclear Nonproliferation Technology and Policy Conference (ANTPC 2018), November 2018, Orlando, FL.
4. B. MOUGINOT, **K. A. MUMMAH**, and P. P. H. WILSON, "Evaluating computational methods for modeling off-normal operation of gas centrifuge cascades," in Proceedings of the 59th Annual Meeting of the Institute of Nuclear Materials Management, Jul. 2018, Baltimore, MD, United States
3. B. MOUGINOT, **K. A. MUMMAH**, and P. P. H. WILSON, "Modeling Potential JCPOA Diversion Scenarios with Cyclus," presented at 2018 American Nuclear Society Annual Meeting, June 2018, Philadelphia, PA.
2. **K. A. MUMMAH**, J. W. BAE, D. O'GRADY, and A. LOPEZ, "NuWWIS: An Interim Storage Solution for Spent Fuel," presented at 2017 American Nuclear Society Student Conference, April 2017, Pittsburgh, PA.
1. **K. A. MUMMAH** and C. KUPRIANCZYK, "Public Image in Spent Fuel Disposal: Lessons Learned from Sweden's SKB," presented at 2016 American Nuclear Society Student Conference, April 2016, Madison, WI, United States.

Technical Reports

3. P. L. LAFRENIERE, **K. A. MUMMAH**, J. TOKEY, C. LURVEY, R. IYENGAR, "Guidance Regarding Probable Approaches and Instrumentation to Liquid Fuel Molten Salt Reactor Material Control and Accounting (MC&A)," Los Alamos National Laboratory (LANL), Dec. 2025, Los Alamos, NM (United States), Tech. Rep. LA-UR-25-31727.
2. **K. A. MUMMAH**, D. E. JACKSON, J. OAKBERG, K. E. APT, and V. HENZL, "Advanced Algorithms for Scrutiny of Mandatory State Reports Declarations to the IAEA (Final Project Report)," Los Alamos

National Laboratory (LANL), Sep. 2023, Los Alamos, NM (United States), Tech. Rep. LA-UR-23-32729. DOI: 10.2172/2208761

1. V. HENZL, **K. A. MUMMAH**, and K. E. APT, "Advanced Algorithms for Scrutiny of Mandatory State Reports Declarations to the IAEA," Los Alamos National Laboratory (LANL), 2022, Los Alamos, NM (United States), Tech. Rep. LA-CP-22-20328.

Posters

5. **K. A. MUMMAH**, "Isotopically modified lithium for fission and fusion reactors", presented at Complex Natural & Engineered Systems capability pillar review, Apr. 2026, LANL
4. **K. A. MUMMAH**, "Good Practices and Clear Communication Gaps in the Effort to Site a Low- and Intermediate-Level Deep Geologic Waste Repository in Ontario," Poster, presented at the Waste Management Symposium, Mar. 2021, Virtual.
3. **K. A. MUMMAH**, B. MOUGINOT, and P. P.H. WILSON, "Off-normal Operation in Gas Centrifuge Enrichment Cascades," presented at the 2018 American Nuclear Society Student Conference Poster Session, Apr. 2018, Gainesville, FL, USA.
2. **K. A. MUMMAH** and R. WILLIAMSON, Investigating the Effects of Fuel Pellet Geometry on Pellet Cladding Mechanical Interaction using BISON," presented at the 2016 American Nuclear Society Winter Meeting, Nov. 2016, Las Vegas, NV, USA.
1. **K. A. MUMMAH**, "Reactivity Management: More than Just Reactor Engineers," presented at the 2015 American Nuclear Society Winter Meeting, Nov. 2015, Washington, D.C., USA.

Service

Los Alamos National Lab

- **Co-chair**, Nuclear Engineering & Nonproliferation division early career group, *Jan. 2025 – Present*

American Nuclear Society (ANS)

- **Member**, ANS, *2013 – Present*
- **Chair**, Decommissioning & Environmental Sciences Division (DESD), *June 2025 – Present*
- Student Conference Judge, *2022 – Present*
- Referee/Reviewer, *2019 – Present*
- **Vice Chair**, DESD, *June 2024 – June 2025*
- **Chair**, Student Sections Committee (SSC), *June 2021 – June 2024*
- **Treasurer**, DESD, *June 2022 – June 2024*
- **Executive Committee**, Fuel Cycle and Waste Management Division (FCWMD), *June 2021 – June 2024*
- **Committee Member**, SSC, *June 2016 – June 2024*
- **Executive Committee**, Young Members Group (YMG), *June 2019 – June 2022*
- **Vice Chair**, SSC, *June 2019 – June 2021*
- **Executive Committee**, Nuclear Nonproliferation Policy Division (NNPD), *June 2018 – June 2021*
- **Committee Member**, Diversity and Inclusion in ANS Committee (DIA), *June 2018 – June 2021*
- **Student Director**, ANS Board of Directors, *June 2017 – June 2019*
- **Committee Member**, Local Sections Committee (LSC), *June 2016 – June 2019*

ANS Trinity Local Section

- **Member**, *May 2022 – Present*
- **Executive Committee**, *June 2025 – Present*

Generation Atomic

- **Advisory Board Member**, *July 2020 – Jan. 2023*

ANS University of Wisconsin-Madison Student Section

- **Member**, *Aug. 2017 – May 2022*
- **Governor**, *Aug. 2018 – May 2019*
- **Advocacy co-chair**, *Jun. 2017 – Nov. 2017*

Rethinking Nuclear/Friends of Fission Midwest Chapter
- **Science Advisor**, *2018 – 2022*

University of Wisconsin–Madison Department of Nuclear Engineering & Engineering Physics
- **Department Representative to the College of Engineering Graduate Advisory Committee**, *Aug. 2018 – Nov. 2022*
- **Chair, Student Advisory Committee**, *2019 – Jan. 2022*

Wisconsin Baja Racing
- **Steering Team Member**, *Aug. 2018 – May 2020*

ANS University of Illinois at Urbana-Champaign Student Section
- **Member**, *Aug. 2013 – May 2017*
- **President**, *Apr. 2015 – Apr. 2016*
- **Internal Vice President**, *Apr. 2014 – Apr. 2015*
- **Freshman President**, *Oct. 2013 – Apr. 2014*

Women in Nuclear, University of Illinois at Urbana-Champaign Student Section
- **Cofounder**, *Aug. 2015*
- **Member**, *Aug. 2015 – May 2017*

Phi Mu, Delta Beta Chapter
- **Academic Excellence Chair**, *Dec. 2016 – May 2017*

University of Illinois Department of Nuclear, Plasma, and Radiological Engineering
- **Undergraduate Advisory Board Representative**, *Apr. 2014 – Dec. 2016*

Engineering Ambassadors, University of Illinois at Urbana-Champaign
- **Ambassador**, *Nov. 2015 – May 2017*
- **Recruitment & Publicity Chair**, *Apr. 2016 – Dec. 2016*

Engineering Council, University of Illinois
- **Student Introduction to Engineering (SITE) Reservations Chair**, *May 2015 – May 2016*

Student Chapter of the American Meteorological Society, University of Illinois
- **Member**, *Jan. 2015 – May 2017*
- **Secretary**, *Apr. 2015 – Apr. 2016*

Illinois Space Society
- **Member**, *Aug. 2013 – Dec. 2014*
- **RASC-AL Technical Team Member**, *Aug. 2014 – Dec. 2014*
- **Representative to Students for the Exploration and Development of Space (SEDS)**, *Aug. 2013 – Sept. 2014*

Awards and Honors

2025: *Nuclear News* 40 Under 40 – 2025

2025: Wisconsin Initiative for Science Literacy (WISL) Award for Communicating PhD Research to the Public

2024-25: Graduate Fellow, **Consortium for Monitoring, Technology, and Verification**

2019-24: Graduate Fellow, **DOE/NNSA Nuclear Nonproliferation International Safeguards (NNIS)**

2022-23: ANS Robert A. Dannels Memorial Scholarship

2022: American Nuclear Society Presidential Citation

2022: Best Presentation, **Nuclear Technology & Society, ANS 2022 Student Conference**

2022: Semi-finalist, **Fulbright U.S. Student Program**

2021-22: ANS Graduate Scholarship
2020-21: ANS Michael J. Lineberry Memorial Scholarship
2020: First place, **Institute of Nuclear Materials Management J.D. Williams Student Paper Award**
2017-19: Graduate Fellow, **Consortium for Verification Technology**
2017-19: ANS Fuel Cycle and Waste Management Division Randall Scholar
2018: Summer Research Fellow, Glenn T. Seaborg Institute
2017: Daniel F. Hang Outstanding Senior Design Award
2015-17: DOE Nuclear Energy University Program IUP Scholarship Award
2015-17: Nuclear Regulatory Commission Scholarship
2015-17: American Nuclear Society Decommissioning & Environmental Sciences Scholarship
2016-17: Roy G. Post Foundation Scholarship
2016-17: Dale W. and Wanda L. Weaver Engineering Scholarship
2016-17: Crowe Horwath Scholarship
2016-17: Edith and Harry Darby Leadership Scholarship
2015-17: Catherine Pritchard Undergraduate Scholarship
2013-17: Mike Harper Leadership Scholarship
2017: Second place, **Nuclear Innovation Bootcamp Pitch Competition**
2016: Alpha Nu Sigma
2016: Best Presentation, **Public Image, ANS 2016 Student Conference**
2016: William R. Schowalter Award
2016: American Nuclear Society Commendation for Service and Leadership
2015-16: Exelon Energy for Education Scholarship Award
2014-16: National Academy for Nuclear Training Scholarship
2014-15: ANS Incoming Sophomore Scholarships
2014: American Nuclear Society UIUC Most Committed Member
2013: University of Illinois James Scholar

Invited Talks

26. **American Nuclear Society**, ANS Members Take on Social Media Webinar, *Panelist*, July 29, 2024
25. **Penn State Engineering Equity Summit/ANS Student**, Radiant Women in Nuclear, *Moderator*, April 6, 2024
24. **Atomic Allies**, Expanding your Outreach Impact for Nuclear Science Week Webinar, *Panelist*, Sept. 13, 2023
23. **American Nuclear Society**, Social Media for Local & Student Sections Webinar, *Panelist*, Aug. 3, 2023
22. **ANS Student Conference**, Saturday Plenary, *Panelist*, Apr. 15, 2023
21. **ANS Student Conference**, Holding Onto a Professional/Personal Identity in Your Career, *Panelist*, Apr. 15, 2022
20. **Women in Nuclear Region III**, Academia Panel, *Panelist*, Aug. 19, 2021
19. **Generation Atomic**, Nuclear Advocacy Lab, *Seminar*, June 17, 2021
18. **American Nuclear Society Trinity Local Section**, *Keynote*, May 21, 2021
17. **Los Alamos National Lab**, Nuclear Engineering and Nonproliferation Division, *Seminar*, May 11, 2021
16. **American Nuclear Society**, What Does Earth Day Mean To You? Webinar, *Panelist*, Apr. 22, 2021
15. **ANS Student**, Interested in Nonproliferation? Opportunities to Build Your Expertise, *Panelist*, Apr. 10, 2021
14. **ANS Student Conference**, Taking Care of You, *Workshop*, Apr. 8, 2021
13. **Purdue University**, Women in Nuclear Student Section, *Seminar*, Mar. 3, 2021
12. **American Nuclear Society**, Graduate Fellowships, *Webinar*, Dec. 21, 2020

11. **NAYGN**, Science Communication Series, *Speaker*, Oct. 13, 2020
10. **UW-Madison**, American Nuclear Society Student Section, *Seminar*, Oct. 8, 2020
9. **Nuclear Energy Agency**, Global Nuclear Science and Engineering Commencement, *Speaker*, Aug. 27, 2020
8. **ANS Annual**, Communicating Safety & Risk to the Public, *Panelist*, June 9, 2020
7. **ANS Young Professionals Congress**, Advocating to Policymakers, *Panelist*, Nov. 16, 2019
6. **Ecomodernism**, Nuclear Big Vs. Small, *Panelist*, Sept. 2019
5. **Idaho National Laboratory**, Atomic Week 2019, *Workshop*, Aug. 6, 2019
4. **ANS Annual**, From Nuclear Advocacy to Policy, From Local to Global, *Panelist*, June 13, 2019
3. **ANS Annual**, Results of the April 2019 "Fastest Path to Zero" Summit and Unconference, *Panelist*, June 12, 2019
2. **ANS Winter**, Advocacy and Communication: A Clean Energy Discussion, *Panelist*, Nov. 1, 2017
1. **Environmental Progress**, Atomic Vintage, *Lightning Talk*, July 26, 2017

Other

Programming languages: Python, C++, MATLAB, R Tidyverse

Tools and software: Cyclus, MCNP, BISON, APAT, Git, LaTeX

Volunteering: Los Alamos Auxiliary Fire Brigade, *Search and Rescue Team Member, Duty Officer, and Alternate Team Leader*. Field Certified in New Mexico (2023), Tracker I (exp. 2028)

Hobbies: Trail running, Backpacking, Nuclear history, Volleyball, Mentoring students